

Quiz (Habanero)

- Q1.** A temperature sensor in a weather station measures temperatures in C . If the sensor readings for 15 consecutive days are added, the sum is 225. What is the average temperature reading if the sum is divisible by 5?
- (A) 15
(B) 15.5
(C) 16
(D) 16.5
(E) 17
- Q2.** What is the remainder when 27 is divided by 9?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 6
- Q3.** A water treatment plant can process 480 gallons of water per hour. If 16 hours are worked, how many gallons of water are treated in total, and is the total divisible by 10?
- (A) 7680, yes
(B) 7680, no
(C) 7681, yes
(D) 7681, no
(E) 7682, yes
- Q4.** What is the remainder when 43 is divided by 5?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 8
- Q5.** If 18 is divided by 3, what is the quotient and is it divisible by 2?
- (A) 6, yes
(B) 6, no
(C) 7, yes
(D) 7, no
(E) 8, yes
- Q6.** A researcher tracks the rainfall in an ecosystem and finds that the total rainfall for 9 consecutive days is 279 mm. Is the average rainfall divisible by 3?
- (A) yes
(B) no
(C) not enough information
(D) maybe
(E) divisible by 9
- Q7.** What is the remainder when 75 is divided by 10?
- (A) 0
(B) 1
(C) 2
(D) 5
(E) 8
- Q8.** A temperature reading of $35^{\circ}F$ is equivalent to how many degrees Celsius, and is the result divisible by 9?
- (A) 1.67, yes
(B) 1.67, no
(C) 2, yes
(D) 2.01, no
(E) 2.22, yes

Open Ended Questions (FRQ)

- Q9.** Perform the division $93 \div 3$ and express the result with a remainder. Show all steps and calculations.
- Q10.** A water tank can hold 2400 gallons of water. If 15% of the tank is filled every hour, how many hours will it take to fill the tank? Use division to solve and show all steps.

Chef's Tips

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- Tip 1: Ensure students understand the concept of remainders in division
- Tip 2: Emphasize the application of divisibility rules for 2, 3, 5, 9, 10
- Tip 3: Encourage students to use real-world examples like weather and environment to practice division